FireLight Single Sign-On – Overview

FireLight Inbound Integrations

Internal Training Materials

5/16/2024

Adam Davis
Technical Writer
Hexure

719-442-6400 (Office)

hexure.com

Copyright © 2024 Hexure, all rights reserved.

Hexure, Insurance Technologies, FireLight® and FireLight® are registered or unregistered trademarks of Insurance Technologies, LLC (IT) in the USA and/or other countries.

ACORD, ACORD ObjX, ACORD OLifE, AL3, ACORD Advantage, ACORD XML, and "Association for Cooperative Operations Research and Development" are registered or unregistered trademarks of ACORD Corporation in the USA and/or other countries.

Microsoft, Microsoft SQL Server, Microsoft Internet Information Server, Windows, and other Microsoft names and logos are either registered or unregistered trademarks of Microsoft Corporation in the U.S.A. and/or other countries.

All other trademarks are the property of their respective owners.

The information contained in this document is current as of the date of the publication. Because Insurance Technologies, LLC must respond to changing market conditions and technology advances, Insurance Technologies, LLC cannot guarantee the accuracy of any information presented after the date of publication.

INSURANCE TECHNOLOGIES, LLC MAKES NO WARRANTIES, EXPRESSED OR IMPLIED, IN THIS DOCUMENT AND HEREBY DISCLAIMS ANY AND ALL SUCH WARRANTIES.

The material contained in this document is considered confidential and the intellectual property of Insurance Technologies, LLC. The recipient is given access to this material on the condition that the recipient (1) will keep the information confidential at all times, and (2) will not copy or modify or share the materials, except as expressly authorized by Insurance Technologies, LLC. The recipient should limit its disclosure of the information and materials only to its employees who have a clear business purpose and need to receive such information and materials and who are bound by confidentiality obligations to the recipient that are at least as protective of such information and materials as those contained herein.

Hexure
Two North Cascade Avenue, Suite 200
Colorado Springs, CO 80903
Phone: 719.442.6400

Internet E-Mail: info@hexure.com
Website: hexure.com

Contents

General Information **3**
- **Typical Scenarios** 3
- **FireLight Data Requirements** 4
  - **User Characteristics** 4
  - **Producer Characteristics** 4

SSO Portal Integrations **4**
- **SAML SSO** 5
  - **SAML SSO Process Diagram** 5
  - **SAML SSO XML Transaction** 5
- **Token SSO** 5
  - **Token SSO Process Diagram** 6
  - **Token SSO XML Transaction** 7
- **Inbound Tx1228 Information** 7
  - **Inbound Tx1228 Data Items** 7
  - **Inbound Tx1228 XML Examples** 10

CONFIDENTIAL ©2024 Hexure. All rights reserved.

Introduction

This document provides an overview of FireLight's "single sign-on" (SSO) implementation and its two types of methods: SAML SSO and Token SSO. Here you'll find information on common SSO scenarios, data requirements, method-specific processes, and 1228 Inbound XML examples. The subject matter below is for any Hexure employee who'd like to learn about FireLight SSO.

Note: In addition to this document, please reference the *Sign On Methods* section of the FireLight Developer Guide as well as the SAML SSO (Solution Consultants) document.

General Information

FireLight is a web application that supports creating and managing activities. Users typically access FireLight through a link on the Carrier web portal, which authenticates users and passes required authorization information to FireLight's system. This login method is called "single sign-on" (SSO).

FireLight differentiates the system's User from a Producer or Agent. The User is the person at the keyboard who's authenticated by the Carrier portal. The Producer, however, determines what products can be sold in FireLight. FireLight requires a unique identifier for the User and Producer, called a *User ID* and *Producer ID*.

Note: In many cases, the User and Producer are the same person.

Typical Scenarios

Scenario	Description
The User is a Producer	• The *User ID* and *Producer ID* are the same. • When the User saves an activity, it appears under their *User/Producer ID*.
A Back Office User is creating an activity for a Producer	• The *User ID* and *Producer ID* are different. • The Back Office User running FireLight has a *User* ID, and the Producer has a *Producer ID*. • Once the Back Office User saves the activity, it appears under the *User ID*. The Producer won't have access to the activity. • The Back Office User must assign the activity to the Producer to transfer ownership and grant them access.

CONFIDENTIAL ©2024 Hexure. All rights reserved.

A Back Office User is signing in on behalf of the Producer	• The Carrier portal registers a Back Office User as a certain Producer. • If the Back Office User saves the activity, it appears under the Producer ID. The Producer will have access to the activity.

FireLight Data Requirements

Business requirements determine the information FireLight needs for SSO. This includes application permissions, data visibility, and sharing capabilities. Typical data needs center around the User and Producer characteristics below.

User Characteristics

1. Authenticated by the Carrier portal.
2. Owns all data in FireLight.
3. Distinguished by a Role Code. For example:
    - Producer/Agent
    - Home Office
    - Sales Desk
    - Wholesaler

Producer Characteristics

1. Full demographic information is typically needed for activity output:
    - First and Last Name, Suffix, and Certifications
    - Agency or Company Name
    - Address, Phone Number(s), and email/web addresses
    - License Information
2. Selling Agreement or Distribution Identifier. This determines products, riders, and other selling privileges.

SSO Portal Integrations

There are two typical integration methods between the Carrier portal and FireLight: SAML SSO and Token SSO (also referred to as REST-based SSO).

- **SAML SSO**: The Carrier portal packages all the information that FireLight requires and posts it to a landing page.
- **Token SSO**: FireLight's API Token Service is used to authenticate users. This method pulls from Inbound Tx1228 XML data when a token is requested.

CONFIDENTIAL ©2024 Hexure. All rights reserved.

SAML SSO

In a SAML SSO method, the Carrier portal gathers all the information that FireLight needs for authentication and authorization and does a "form post" to a FireLight landing page. The authentication information that the portal gathers is contained in a payload, which is in XML format.

Several security layers can be added to a payload in order to: 1) prevent tampering; 2) ensure that the payload comes from the Carrier portal.

SAML SSO Process

1. The User initiates the log in process through the Carrier portal link.
2. The Carrier portal submits the login information (in an XML format) through a standard HTTP form post operation.
3. FireLight processes and validates the XML.
4. FireLight executes one of the following actions:
    - Creates the User and/or Producer if they don't exist in FireLight.
    - Updates the User/Producer information if they already exist in the application.
5. Once the payload processing is complete, the User is taken to the FireLight Home page.

SAML SSO XML Transaction

FireLight can use an ACORD Inbound Tx1228 for the post over, which contains User and Producer information. The Tx1228 is usually custom for each Carrier implementation, depending on their business requirements.

For information on the Inbound Tx1228's data types, and an example of the transaction's XML, reference this document's Inbound Tx1228 Information section.

Token SSO

In a Token SSO method, the User chooses a product and jurisdiction within the Carrier portal and begins a FireLight session. Then the client's web service sends authentication information to FireLight's base API token service, and in return, the web service receives a web token that validates the user.

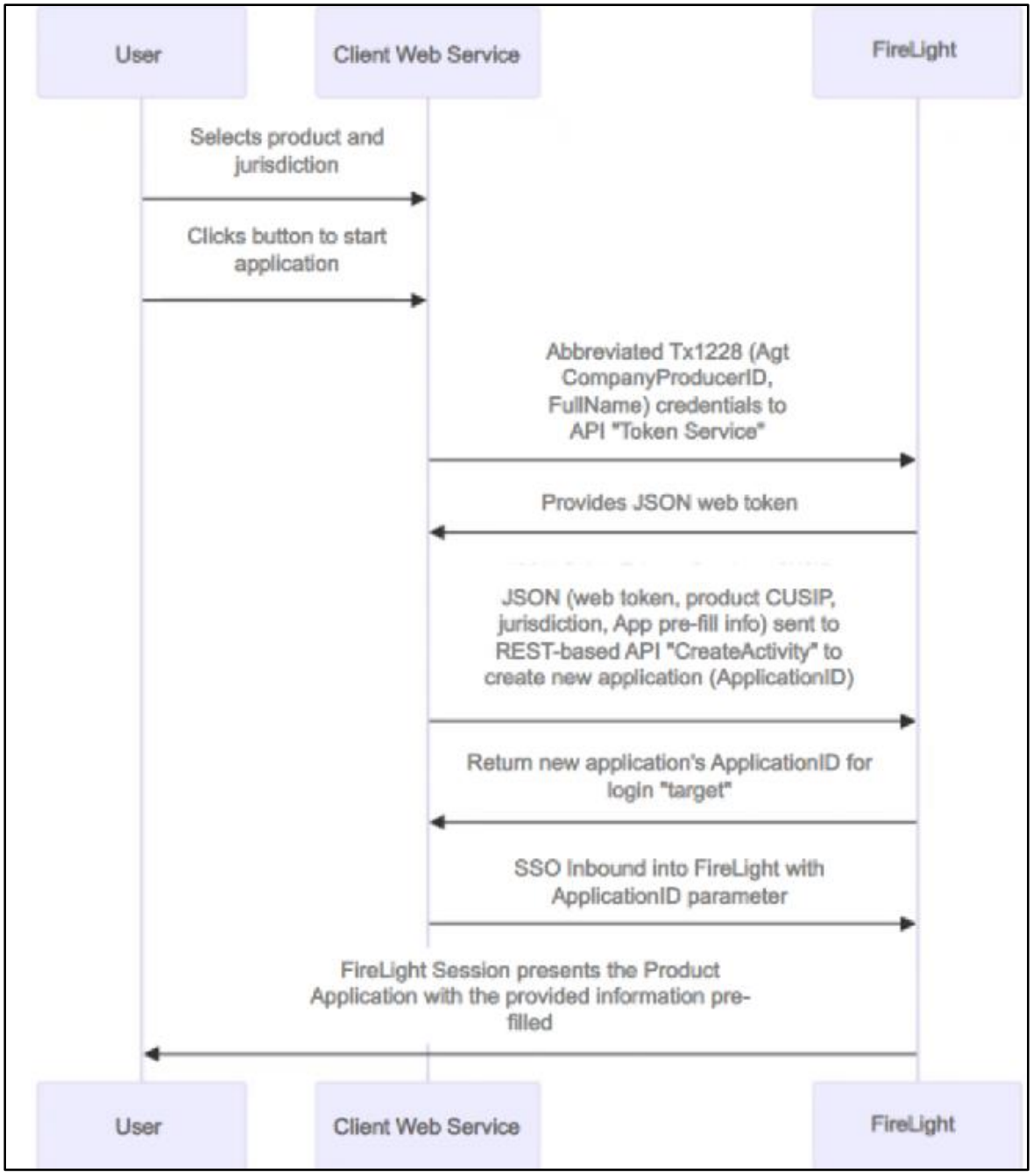
FireLight uses the token and authentication information to do the following:

1. Verify the User.

CONFIDENTIAL ©2024 Hexure. All rights reserved.

2. Prepopulate the User's application (if the client sends in data about the application via an abbreviated Tx1228 payload).
3. Direct the User to the application within FireLight.

Token SSO Process Diagram



CONFIDENTIAL ©2024 Hexure. All rights reserved.

The following steps occur within this process:

1. The user chooses the product and jurisdiction within the client's portal.
2. Within the client's portal, the user starts the FireLight session.
3. The client's web service sends an abbreviated 1228, containing the *Company Producer ID* and *FullName*, to the base API token service.
4. The base API token service responds with a web token that validates the user. **Note**: This token is valid for one hour.
5. The client's web service uses the web token and sends in the application's CUSIP and jurisdiction through a JSON request. The client can also send in additional data about the application for prepopulating purposes if they desire.
    a. The JSON request is sent from the client's web service to Hexure's REST-based API (*CreateActivity*). The request triggers an application's creation for its CUSIP and jurisdiction.
6. The REST-based API sends back the new application's GUID to the client's web service.
7. The client's web service performs a SAML inbound with the GUID.
8. The user logs into FireLight and lands on the application, which has prepopulated information, if the client sent additional data about the application.

Token SSO XML Transaction

Like the SAML SSO method, FireLight uses an ACORD Inbound Tx1228 for Token SSO, which contains User and Producer information. The Tx1228 is usually custom for each Carrier implementation, depending on their business requirements.

For information on the Inbound Tx1228's data types, and an example of the transaction's XML, reference this document's Inbound Tx1228 Information section below.

Inbound Tx1228 Information

This section outlines the data items that can be in an Inbound Tx1228 and provides an example of the 1228's XML structure.

Inbound Tx1228 Data Items

Here's the data that can be captured in an Inbound Tx1228 (the data below may be limited depending on what the Carrier can provide):

<UserAuthRequest>

The *<UserAuthRequest>* aggregate contains login information that FireLight uses to authenticate the external system. This is required when the Tx1228 is a POST to the passive FireLight endpoint.

CONFIDENTIAL ©2024 Hexure. All rights reserved.

Component	Description	Required or Optional
<UserLoginName>	FireLight login name. All requests use the same login name and password.	Required
<UserPwd>	The password for authenticating the external system with FireLight.	Required
<CryptType>	Required with a value of "AES".	Required
<CryptPswd>	Contains the shared password's SHA1 hash, which is appended to the aggregate's SHA1 hash. It's signed using "AES" and the entire value base64 encoded.	Required
<UserDomain>	Enables FireLight to identify the organization subscription.	Required
<VendorApp>	An external system's identifier.	Required
<AppName>	An external system's name.	Required

<TxLifeRequest>

The *<TxLifeRequest>* aggregate holds the FireLight user's data. It's required. *<TxLifeRequest>* can be comprised of the following components:

Component	Description	Required or Optional
<TransRefGUID>	Randomly generated, alphanumeric string. This must be unique for each transaction.	Required
<TransType>	The "tc" value must be **1228**.	Required
<TransSubType>	The "tc" value must be **22800**.	Optional
<OLifE>	An aggregate that's nested below <TxLifeRequest>. It contains the data in this table's following rows.	Required
<SourceInfo>	Indicates the actual referring system identifier. For example, if your client has multiple ways to log into FireLight, you can use <SourceInfoName> to identify which launch mechanism was used.	Optional
<Party>	Identifies the system user. You can also use this aggregate to identify Firms that	Required

CONFIDENTIAL ©2024 Hexure. All rights reserved.

	the user is associated with (this is optional).	
<FullName>	Optional if <Person>, <FirstName>, and <LastName> are provided. Required if <Person> aggregate isn't used.	Required or Optional
<Person>	Required if <FullName> isn't included within the aggregate. **Note**: When the <NickName> attribute is used with the <Person> aggregate, it allows a user to log into FireLight under another user's account and maintain accurate case ownership information.	Required or Optional
<Producer>	Identifies the user within FireLight's system.	Required
<CarrierAppointment>	Identifies the user and/or firm's distribution agreement.	Required
<CompanyProducerID>	Unique identifier from the external system that FireLight uses to identify the user.	Required
<CarrierCode>	Identifies the organization subscription to FireLight.	Required
<Relation>	Identifies the aggregate's role.	Required
<DistributionAgreementInfo>	Identifies the distribution agreement aggregate for the organization.	Optional
<Registration>	Identifies the organization firm information.	Optional
<FirstName>	Identifies the organization firm name.	Optional
<FirmRegistrationNum>	Identifies the organization Firm ID.	Optional
<Relation>	Identifies the <Party> aggregate's role and the relation between Person and Organization.	Required

<DistributionAgreement>

The <DistributionAgreement> aggregate is used when specifying distribution agreements and associated firms.

CONFIDENTIAL ©2024 Hexure. All rights reserved.

Component	Description	Required or Optional
<DistributionAgreement>	Identifies product availability.	Optional
<PolicyProductInfo>	Identifies a product defined in FireLight.	Optional
<ProductCode>	External code to help identify a product within FireLight.	Optional
<OLifEExtension>	Extends the ACORD (v2.27) model.	Optional
<CusipNum>	External code to help identify a product defined in FireLight.	Optional
<DistributionAgreement>	Identifies product availability.	Optional

Inbound Tx1228 XML Examples

FireLight can use two types of Inbound Tx1228 XML payloads: a payload for Product Filtering, and a payload for Distribution Agreements (also referred to as Firm IDs).

- **Product Filtering**: The Carrier's web portal determines what products the Producer can see.
- **Distribution Agreements**: A formal contract between a Carrier and Distributor that specifies what products can be sold within FireLight and who can sell them. An Inbound Tx1228 XML payload can carry the Distribution Agreement ID, Policy Product Info ID, Product Code, and Carrier Code.
    - **Note to FireLight Solution Consultants**: If your client's implementation contains a Distribution Agreement, then be sure to include data items that are related to Firm IDs in their Inbound Tx1228 mapping document. For more information on this, please reference the Mapping an Inbound 1228 document as well as this document's XML Transaction (with Distribution Agreement) section.

CONFIDENTIAL ©2024 Hexure. All rights reserved.

XML Transaction (with Product Filtering)


```xml
<TXLife xmlns="http://ACORD.org/Standards/Life/2">
  <TXLifeRequest>
    <TransRefGUID>32641092-94b1-4b72-b79c-2fab56e77aa1</TransRefGUID>
    <TransType tc="1228"/>
    <TransSubType tc="22800"/>
    <OLifE>
      <SourceInfo>
        <CreationDate>2016-10-11</CreationDate>
        <CreationTime>10:47:14-07:00</CreationTime>
        <SourceInfoName>%CLIENT_ACRONYM%</SourceInfoName>
      </SourceInfo>
      <Party id="Agent_Party">
        <PartyTypeCode tc="1">Person</PartyTypeCode>
        <!--  Agent Name  -->
        <FullName>Sample T User</FullName>
        <!--  Plain SSN  -->
        <GovtID>999-99-9999</GovtID>
        <Person>
          <FirstName>Sample</FirstName>
          <MiddleName>T</MiddleName>
          <LastName>User</LastName>
          <!--  Date of Birth  -->
          <BirthDate>1985-08-25</BirthDate>
        </Person>
        <!--  Home Address  -->
        <Address>
          <AddressTypeCode tc="1">OLI_ADTYPE_HOME</AddressTypeCode>
          <Line1>123 Home Sweet St</Line1>
          <Line2>Apt A</Line2>
          <City>Bedford</City>
          <AddressStateTC tc="21">OLI_USA_KY</AddressStateTC>
          <Zip>40006</Zip>
          <AddressCountryTC tc="1">OLI_NATION_USA</AddressCountryTC>
        </Address>
        <!--  Business Phone  -->
        <Phone>
          <PhoneTypeCode tc="2">OLI_PHONETYPE_BUS</PhoneTypeCode>
          <AreaCode>(502)</AreaCode>
          <DialNumber>555-7864</DialNumber>
        </Phone>
        <!--  Home Phone  -->
```


CONFIDENTIAL ©2024 Hexure. All rights reserved.

```
        ▼<Phone>
            <PhoneTypeCode tc="1">OLI_PHONETYPE_HOME</PhoneTypeCode>
            <AreaCode>(502)</AreaCode>
            <DialNumber>555-7864</DialNumber>
          </Phone>
          <!--  Fax Number  -->
        ▼<Phone>
            <PhoneTypeCode tc="19">OLI_PHONETYPE_FAX</PhoneTypeCode>
            <AreaCode>(502)</AreaCode>
            <DialNumber>555-7865</DialNumber>
          </Phone>
          <!--  Producer  -->
        ▼<Producer>
          ▼<CarrierAppointment>
              <!--  Firelight uses the following to uniquly identify the user log-in
              <CompanyProducerID>1223456789</CompanyProducerID>
              <!--  3-Digit ID  -->
              <CarrierCode>%CLIENT_ACRONYM%</CarrierCode>
              <!--  List of Products Available to user log-in - 1 Node per Product
            ▼<LineOfAuthority>
                <!--  Product Type code and Description per ACORD  -->
                <LineOfAuthorityType tc="6">Annuity-Fixed</LineOfAuthorityType>
              ▼<OLifEExtension VendorCode="25" ExtensionCode="AvailableProduct">
                  <!--  Product CUSIP  -->
                  <AvailableProduct>34628V631</AvailableProduct>
                </OLifEExtension>
              </LineOfAuthority>
            </CarrierAppointment>
          </Producer>
          <!--  Email Address  -->
        ▼<EMailAddress>
            <AddrLine>sampleuser@example.com</AddrLine>
          </EMailAddress>
        </Party>
        <!--  Role Code (defined through Firelight Admin)  -->
      ▼<OLifEExtension VendorCode="25">
          <UserRoleCode tc="%CLIENT_ACRONYM%_Agent">Agent</UserRoleCode>
        </OLifEExtension>
      </OLifE>
    </TXLifeRequest>
  </TXLife>
```


CONFIDENTIAL ©2024 Hexure. All rights reserved.

XML Transaction (with Distribution Agreement)


```
▼<TXLife xmlns="http://ACORD.org/Standards/Life/2">
 ▼<TXLifeRequest>
    <TransRefGUID>26C1A4FC-C22C-4027-9BF6-575AF38AAE2A</TransRefGUID>
    <TransType tc="1228">OLI_TRANS_TRNPRODINQ</TransType>
    <TransSubType tc="22800">OLI_TRANSSUB_PRODDETALL</TransSubType>
   ▼<OLifE>
    ▼<Party id="Agent_Party">
       <PartyTypeCode tc="1">OLI_PT_Person</PartyTypeCode>
       <FullName>Agents B</FullName>
      ▼<EMailAddress>
         <AddrLine>agentB_email@insurancetechnologies.com</AddrLine>
       </EMailAddress>
       <Person/>
      ▼<Producer>
       ▼<CarrierAppointment PartyID="Agent_Party">
          <CompanyProducerID>TEST_AGENT</CompanyProducerID>
          <CarrierCode>XYZ</CarrierCode>
        </CarrierAppointment>
      </Producer>
     </Party>
    ▼<Party id="Firm_Party_1">
       <PartyTypeCode tc="2">Organization</PartyTypeCode>
       <FullName>Firm ABC</FullName>
      ▼<Producer>
       ▼<CarrierAppointment>
          <CarrierCode>XYZ</CarrierCode>
          <DistributionAgreementInfo id="DistributionAgreementID_1ID" DistributionAgreementID="[
        </CarrierAppointment>
       ▼<Registration>
          <FirmName>Firm ABC</FirmName>
          <FirmRegistrationNum>FirmABC</FirmRegistrationNum>
        </Registration>
       </Producer>
      </Party>
     ▼<Party id="Firm_Party_2">
       <PartyTypeCode tc="2">Organization</PartyTypeCode>
       <FullName>Firm XYZ</FullName>
      ▼<Producer>
       ▼<CarrierAppointment>
          <CarrierCode>XYZ</CarrierCode>
          <DistributionAgreementInfo id="DistributionAgreementID_2ID" DistributionAgreementID=
        </CarrierAppointment>
       ▼<Registration>
          <FirmName>Firm XYZ</FirmName>
          <FirmRegistrationNum>FirmXYZ</FirmRegistrationNum>
        </Registration>
       </Producer>
      </Party>
     ▼<Relation OriginatingObjectID="Agent_Party">
       <RelationRoleCode tc="11">OLI_REL_AGENT</RelationRoleCode>
      </Relation>
```


CONFIDENTIAL ©2024 Hexure. All rights reserved.

```
▼<Relation OriginatingObjectID="Firm_Party_1" RelatedObjectID="Agent_Party">
   <RelationRoleCode tc="120">Agent of Agency</RelationRoleCode>
   <Sequence>0</Sequence>
 </Relation>
▼<Relation OriginatingObjectID="Firm_Party_2" RelatedObjectID="Agent_Party">
   <RelationRoleCode tc="120">Agent of Agency</RelationRoleCode>
   <Sequence>1</Sequence>
 </Relation>
▼<DistributionAgreement id="DistributionAgreementID_1">
  ▼<PolicyProductInfo id="PolicyProductInfo_1">
     <ProductCode>ASHARE</ProductCode>
     <CarrierCode>XYZ</CarrierCode>
    ▼<OLifEExtension>
       <CusipNum>12345</CusipNum>
     </OLifEExtension>
   </PolicyProductInfo>
 </DistributionAgreement>
▼<DistributionAgreement id="DistributionAgreementID_2">
  ▼<PolicyProductInfo id="PolicyProductInfo_3">
     <ProductCode>XSHARE</ProductCode>
     <CarrierCode>XYZ</CarrierCode>
    ▼<OLifEExtension>
       <CusipNum>54321</CusipNum>
     </OLifEExtension>
   </PolicyProductInfo>
 </DistributionAgreement>
</OLifE>
</TXLifeRequest>
</TXLife>
```


CONFIDENTIAL ©2024 Hexure. All rights reserved.